



Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.

UNIT 2 • STANDARD 6.DS.4

Describe Data Using the Mean

Lesson 2-8 • Teacher Copy

Name: _____

Date: _____

Class: _____



TODAY'S OBJECTIVES

Content Objective: I can determine the mean of a data set, use a target mean to find a missing value, and explain the mean as the fair share and the balance point of the data.

Language Objective: I can explain my reasoning using the words mean, average, measure of center, balance point, and data set.



See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.



Fill in each blank as we go. Use the Word Bank to help you.



WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Mean

Average

Measure of center

Balance point

Data set



Tap any word to see what it means and a picture.

Level 3: try the blanks from memory first, then check the bank.

1

The value found by adding all the data and dividing by how many values there are is the .

2

A common everyday word that means the same thing as the mean is the .

3

A single number that describes the typical value of a whole data set is a .

4

The value where the total amount above equals the total amount below is the .

5

The whole collection of collected values is the .

Write About the Math

How many points does Sofia need to score in the last game to hit her goal, and how does the balance-point view give the same answer as adding and dividing?

¿Cuántos puntos necesita anotar Sofía en el último partido para llegar a su meta, y cómo da la vista del punto de equilibrio la misma respuesta que sumar y dividir?

Say your answer to a partner first. Then write it, using at least two of these words:

☐ **Mean**
Media

☐ **Average**
Promedio

☐ **Measure of center**
Medida de centro

☐ **Balance point**
Punto de equilibrio

☐ **Data set**
Conjunto de datos

Model: A typical day is the amount Miguel would earn if he earned the SAME amount every day. — Students say a typical day is the equal-share amount, and that sharing \$60 across 4 days means dividing.

Start

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- After 9 games Sofia has scored ____ points. To average 15 across 10 games she needs ____ points in all, so game 10 must be ____ points. Using the balance point instead, her games total ____ points below the target mean and ____ points above it, so game 10 has to land ____ the target mean.
 - My answer is ____ because ____.
Mi respuesta es ____ porque ____.
-

Build

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- My claim is ____.

Mi afirmación es ____.

- **My evidence is** ____, **so** ____.

Mi evidencia es ____, entonces ____.

Explain

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

☐ I answered the question.

☐ I used math evidence: numbers, an equation, or a model.

☐ I used at least two math words.

Writing Guide — Teacher Copy

- The answer to the math question is correct.
- The evidence is real lesson mathematics (numbers, equation, or model), not restated claim.
- The support level changed the language scaffolding, never the mathematical expectation.

Answer Key & Teacher Guide

1. **Notes 1:** Mean
2. **Notes 2:** Average
3. **Notes 3:** Measure of center
4. **Notes 4:** Balance point
5. **Notes 5:** Data set
6. **Watch for this mistake:** *The most common mistake with the mean is adding correctly and then dividing by the wrong number — or not dividing at all. Students report the total (for 8, 10, 6, 12 they say the mean is 36) or they divide by a count that does not match the data, such as dividing a 4-value total by 5, or skipping a zero when counting the values. Before you finish, do two things: count the values in the data set out loud, and check that your mean lands between the smallest and largest value. A mean of 36 for data that never goes above 12 is impossible.*
7. **Try It 1:** A. 17 points — *She needs 150 points in all and already has 133, so game 10 must supply $150 - 133 = 17$ points.*
8. **Try It 2:** A. 24 — *If every value moves up by 4, the balance point moves up by 4 as well, so the new mean is $20 + 4 = 24$. In totals: adding 4 to each of n values adds $4n$ to the total, and $4n \div n = 4$.*